Generative AI Analysis and Ethics Report

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Overview

I train a generative model that makes new 64 by 64 colour images of British and Irish butterfly specimens, pinned and photographed as the Natural History Museum (NHM), London, photographs them. My dataset comprises 22,000 specimen photographs from iCollections, the digitisation of British and Irish butterflies (Paterson et al., 2016).

The model type, from the brief's list, is a "Generative Adversarial Network (GAN) for image generation". A GAN trains two networks against each other (Goodfellow et al., 2014, p. 1). A generator turns random noise into an image, and a discriminator gives each image a score, higher for one it takes as real. Mine is an unconditional deep convolutional GAN (DCGAN) at 64 by 64 pixels, so it takes no species label. I wrote and trained it from scratch in the notebook, with no pre-trained network.

The generator learned to make recognisable pinned butterflies on a museum tray. Of 64 numbered samples, 24 score full marks on my written rubric, and three score under 3 of 5. My memorisation check measures how close each sample is to its nearest training image. Its line is a rule of thumb, set from how close real photographs kept out of training come to the training images, so I call it a heuristic threshold. None of the 64 samples crossed that heuristic threshold, but the check cannot rule out memorisation in other forms. Two parts did not go to plan, and I report both. The checkpoint rule I wrote before training misfired and picked a near-untrained model, so the model I evaluate was chosen after I had seen the outputs. I also tested one real-world use, the trained discriminator as a quality flag for museum photographs, and in that exploratory test it did not work.

Dataset or Prompt Description

The Data. The photographs come from the Specimens resource of the NHM Data Portal's Collection specimens data set (Natural History Museum [NHM], n.d.-b). I selected the iCollections butterflies: order Lepidoptera (moths and butterflies), project iCollections, and the six butterfly families, one query each. That project digitised "181,545 specimens representing 89 species from the British Isles and Ireland" (Paterson et al., 2016, p. 1). Its protocol explains what each photograph holds: "Specimens are photographed in a unit tray specially constructed to allow the specimens to be imaged together with their labels." (Paterson et al., 2016, p. 4). So each photograph shows a ruler, one pinned butterfly and its handwritten or printed labels.

Licence. The data set's own record gives its specimen records a Creative Commons Zero (CC0) licence (NHM, n.d.-b). The photographs carry a licence of their own. The Museum's citation page states that "All images in the Specimens resource are shared under a Creative Commons Attribution 4.0 International (CC BY 4.0) License" (NHM, n.d.-a). My download script kept a photograph only if its own licence field gave CC BY 4.0, and all 22,000 do. In the notebook I credit the six photographs of its first figure in the form that page asks for. The project's attribution file credits all ten photographs of the notebook's first two figures in that form. Where a figure here shows the Museum's photographs, I credit the Museum and link the licence in its note. The Museum cannot warrant the quality or accuracy of the data, and it has not endorsed this work.

Size and Scope. The script fetched the 22,000 photographs on 13 October 2026, one per specimen, in a seeded random order. It left out 2,635 records with more than one specimen photograph. Most photographs are 1,440 pixels wide: 14,543 portrait and 7,292 landscape. The other 165 are the Museum's 512-pixel rendition. Lycaenidae (the blues, coppers and hairstreaks)

make up 42.7 per cent of the set and Nymphalidae (the browns, fritillaries, admirals and tortoiseshells) 38.7 per cent. Pieridae (the whites and yellows) follow at 12.3 per cent, Hesperiidae (the skippers) at 4.8, Riodinidae (here only the Duke of Burgundy) at 1.0 and Papilionidae (the swallowtails) at 0.5. By country, 86.1 per cent come from the United Kingdom (UK), 4.4 per cent from Ireland and 0.1 per cent from the Isle of Man. The other 9.4 per cent have no country, so I never call the set "UK only".

How It Supports the Task. The set holds thousands of photographs of one kind of object, taken to one protocol. A GAN learns the distribution of the images it is shown, and I expect uniform framing to make that distribution easier to learn. I held out a seeded 10 per cent of the specimens, 2,200 photographs, which never trained the model or chose a model setting. That leaves 19,800 for training. A fixed manifest lists every photograph with its hash, and my download script rebuilds the set from it, so the project can be rerun.

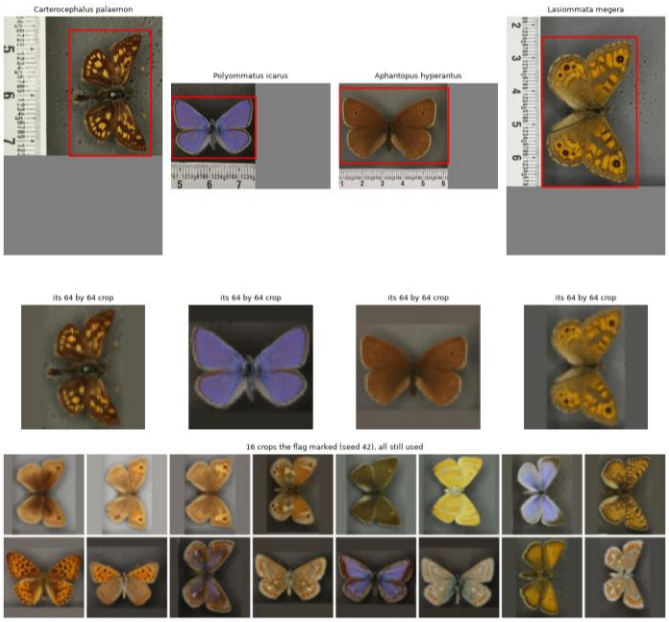
How I Prepared It. A crop rule finds the ruler and the label section, marks the pixels that differ from the tray's grey, and boxes the specimen with a small margin. Each box is padded to a square with the tray's colour, so the specimen keeps its shape, and resized to 64 by 64 pixels. I fixed the rule before any training and used it for every run. In comparing generative models, "changes in preprocessing, even very small and subtle ones, are completely unacceptable" (Goodfellow et al., 2016, p. 714). No source I hold covers the crop rule itself, so I measured it instead.

Before training, 600 crops were scored by eye on contact sheets (grids of crops with their boxes drawn), with an AI assistant doing the scoring. I reviewed the results. The verdicts ship with the notebook as a file, one row per crop, and the notebook counts every rate below from it. These 600 were drawn from the whole set, so they include both training and held-out

photographs: 545 training and 55 held-out. The audit is a different selection from the held-out split. The rule failed on 10 of the 200 images I set the rule on (5.0 per cent) and on 24 of 400 further images (6.0 per cent), so about 6 per cent in all. Of those 200, 21 are held-out photographs. Of the 600, 63 were marked doubtful, borderline crops that I count with the passes, so they are not in the failure rate. Of the 34 failures, 21 kept part of a label or card and 13 cut into the specimen. I also built an automatic crop flag from the rule's own signals. It caught 9 of the 10 failures in the 200 it was set on, but only 9 of the 24 in the other 400, where it also raised 62 false alarms. It marks 4,233 of the 22,000 crops, 19.2 per cent. Figure 1 shows the rule on four training photographs, and 16 crops the flag marked. Filtering by it would drop about one photograph in five, most of them good, and keep 15 of the 24 failures in the 400 it was not set on. It would also shift the family shares, because it marks 41.6 per cent of swallowtails and 8.8 per cent of skippers. So I train on every crop and report the flag as a finding.

Figure 1

The Crop Rule on Four Training Photographs, Their Crops, and 16 Flagged Crops



Note. From notebook Figure 2. The red box is the crop, and grey paint hides the label section. Photographs adapted from the Natural History Museum (n.d.-b), CC BY 4.0 (https://creativecommons.org/licenses/by/4.0).

I listed every anomaly in the data before training. For a problem value, "The options are limited to correcting, deleting, or leaving unchanged." (Van den Broeck et al., 2005, p. 968). I left every row unchanged and marked it. The set holds no duplicate file and no duplicate specimen. Of the photographs, 2,064 have no country, 196 carry a hybrid name and 15 have no species. The model takes no names or countries as input, but these photographs stay in training and shape what it learns, as every photograph does. Last, each batch is scaled to the range -1 to 1. That matches the DCGAN paper, where "No pre-processing was applied to training images besides scaling to the range of the tanh activation function [-1, 1]." (Radford et al., 2016, p. 3). Training also flips each image left to right at random, a step from my coursework that I kept as

my own choice. Somepalli et al. (2023, p. 6052), in a study of diffusion models, also "train all models with random horizontal flip and random crop augmentations".

In plain terms, the model sees a small, square, low-resolution picture of each pinned butterfly, with the ruler and most of the labels cut away.

Model Design and Training Approach

Why a GAN. The brief pairs a GAN with image generation. A variational autoencoder was the other image route, but "samples from variational autoencoders trained on images tend to be somewhat blurry" (Goodfellow et al., 2016, p. 694). Blur would hide the failure cases I had to name. I found no peer-reviewed GAN study on museum specimen photographs that I could read at the publisher. So I based the choice on the general GAN design papers, not on a study of this kind of image.

The Two Networks. The two networks and the training loop are adapted from my coursework DCGAN on the CelebA faces, and the notebook's Task 3 lists each place I depart from it. The generator takes a noise vector of 100 random numbers. Five transposed convolutions take it from 1 by 1 to 64 by 64 pixels, with 64 base channels. The first four are followed by batch normalisation (BatchNorm) and a rectified linear unit (ReLU), and the last by the hyperbolic tangent (Tanh). It has 3,576,704 parameters. The DCGAN rule is to "Use ReLU activation in generator for all layers except for the output, which uses Tanh." (Radford et al., 2016, p. 3). The discriminator mirrors it with five convolutions, the first four strided, from 64 by 64 pixels to one score, and has 2,765,568 parameters. Its output runs from 0 to 1, but with smoothed targets it is not a calibrated probability that an image is real, so I call it the discriminator's score. Its design follows the rule "Replace any pooling layers with strided convolutions (discriminator) and fractional-strided convolutions (generator)." (Radford et al., 2016, p. 3). It uses leaky ReLU with

a slope of 0.2, as "In the LeakyReLU, the slope of the leak was set to 0.2 in all models." (Radford et al., 2016, p. 3). Weights start from a normal distribution with standard deviation 0.02, as Radford et al. (2016, p. 3) did. Before training, the generator gives a near-flat grey, with pixel values from 125 to 130.

The Loss. Both networks use binary cross-entropy (BCE). The discriminator's loss is the sum of its real and generated terms. A discriminator that gives 0.5 to every image has a summed loss of 2 ln 2, which is 1.386, and I draw that as the constant-0.5 reference. My check on the untrained discriminator gave 1.3857. Real images get a target of 0.9 and generated ones 0. With these targets, a discriminator that cannot tell real from generated does best by giving every image 0.45, where the summed loss is about 1.376, just under the reference. This is one-sided label smoothing: Salimans et al. (2016, p. 4) "smooth only the positive labels to" a value below 1. They show that a non-zero target for generated images is "problematic" (Salimans et al., 2016, p. 3). The generator's target is 1, the non-saturating loss, which "provides much stronger gradients early in learning" (Goodfellow et al., 2014, p. 3).

The Optimiser and Schedule. Each network uses Adam at a learning rate of 0.0002, with betas of 0.5 and 0.999. Radford et al. (2016, p. 3) "found the suggested learning rate of 0.001, to be too high, using 0.0002 instead", and for the first beta, "reducing it to 0.5 helped stabilize training" (p. 4). Batches hold 128 images, the DCGAN paper's "mini-batch size of 128" (Radford et al., 2016, p. 3). That gives 154 full batches an epoch, one shuffled pass through the training images, and 15,400 generator updates in 100 epochs. The 100 epochs are my own assumption, guarded by the checkpoint rule below. I keep an exponential moving average (EMA) of the generator's weights, with a decay of 0.999, and it renders every sample. That follows my coursework DCGAN and is my own choice, as no source I hold covers it. The EMA

starts as a copy of the untrained generator, with no warm-up. Training runs from seed 42, in 32-bit floating point with deterministic kernels, so a rerun on the same machine should repeat its numbers.

The Small-Data Risk. "Training generative adversarial networks (GAN) using too little data typically leads to discriminator overfitting, causing training to diverge." (Karras et al., 2020, p. 12104). With 19,800 training images, my generator still has far more parameters than images. GANs trained on small sets "can generate almost exact replication of training data" (Feng et al., 2021, p. 6704). So the memorisation check in the next section is my evidence on copying, within the limits I set out there.

The Checkpoint Rule and Its Misfire. A GAN has no validation loss, and GANs "lack an objective function, which makes it difficult to compare performance of different models" (Salimans et al., 2016, p. 4). So before training I wrote a rule to pick the final model: the EMA generator at the last epoch, unless a failure flag fires. Five flags read training-side numbers only. F1 and F2 catch a discriminator that dominates or fails, and F3 an unstable generator loss. F4 catches mode collapse, a generator that makes only a few kinds of image, and F5 a value that is not finite. F1 fires when the epoch-mean discriminator loss is below 0.425 for 5 epochs in a row, and F2 when it is above 1.5 for 5 epochs in a row, after epoch 5. F3 fires when the standard deviation of the generator loss over the last 10 epochs is above 1.0, from epoch 10 on, so its window can reach back to epoch 1. F4 fires at a checkpoint from epoch 10 on when the diversity ratio falls below 0.5. That ratio is the mean pixel distance among the 64 images the EMA generator makes from a fixed batch of noise, divided by the same distance among 64 training images. So 1.0 is as varied as real photographs. Checkpoints fall at epochs 1, 5 and 10, then every 10 epochs to 100. Any flag but F2 in run 1 called run 2, which adds spectral normalisation

to the discriminator (Miyato et al., 2018, p. 1). The same flags in run 2 called run 3, which also slows the discriminator to 0.00005, the two time-scale update rule (TTUR) of Heusel et al. (2017, p. 1). F2 called no further run, and there was no fourth run. The final model came from the last run made. If a flag fired, the model fell back to the last checkpoint from epoch 5 on before the flag's window began. If several flags fired, the earliest window decided, and a run with no such checkpoint had failed.

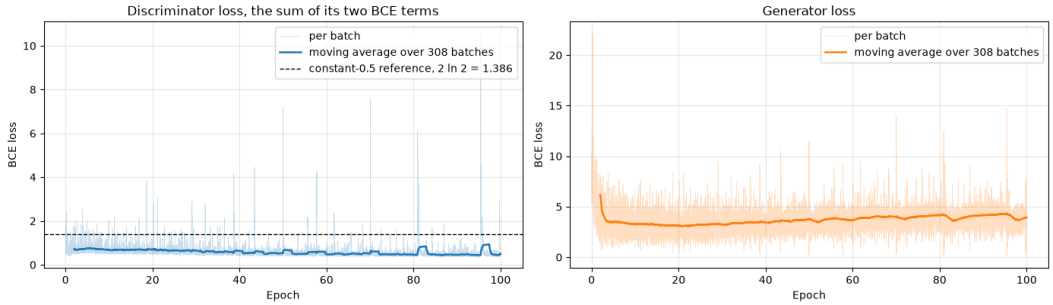
The rule misfired. In run 1, F3 fired at epoch 10, with its window reaching back to epoch 1, so run 1 had no checkpoint to fall back to and failed. Run 2 failed the same way, with F3 and F4 at epoch 10. In run 3, only F4 fired, at epoch 10, and run 3 was the last run made. The rule therefore picked run 3's EMA generator at epoch 5, the last checkpoint before F4's window. Its samples are fine noise textures, with a diversity ratio of 0.07, so it is close to untrained, and at epoch 5 the EMA still holds about half of its random start. I then tried one correction: no flag window takes in epoch 1. It did not clear the runs. F3 then fires at epoch 11, because run 1's generator loss spread sits near the line all through training. Of its 91 ten-epoch windows, 46 are at or under the line, and the last is 1.032. F4 read the EMA generator, which at epoch 10 still held about a fifth of its random start, while the live generators, the ones being trained, already had ratios of 0.90 to 1.04. So F4 caught the EMA's warm-up lag, not a collapse. From epoch 20 on, no run's ratio fell below 0.86. So F3's line was set too tight for this model, and F4 read the EMA before it had warmed up. The generator itself trained well. I did not re-tune the rule after seeing the results, because that would fit the rule to the answer. A future version would read F4 on the live generator, or warm the EMA up, and would set F3's line from a trial run before training.

The model I evaluate is run 1's EMA generator at epoch 100. It is the plain recipe I set before training, from the first run, at its last epoch, so it takes the least choosing. I chose it after seeing the outputs of all three runs, so the choice is not blind. I confirmed it on 14 October 2026, and runs 2 and 3 stay in the notebook as evidence on stability.

How Training Behaved. Run 1 took 6.5 minutes on an NVIDIA GeForce RTX 5090, about 3.9 seconds an epoch. PyTorch's peak was 951 mebibytes (MiB) allocated. Figure 2 shows run 1's losses for every batch, with moving averages.

Figure 2

Run 1's Discriminator and Generator Losses for Every Batch



Note. From notebook Figure 5. Faint lines are single batches and solid lines their moving averages. The dashed line is the constant-0.5 reference, 2 ln 2 = 1.386, the summed loss of a discriminator that gives every image 0.5.

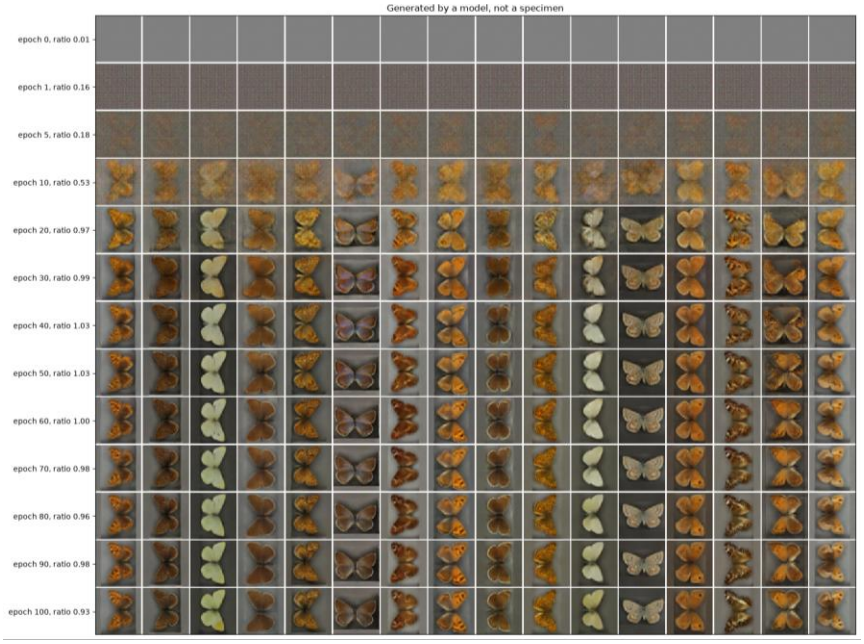
The discriminator led from the start and kept leading. Its loss sat under the constant-0.5 reference almost throughout, falling from 0.676 in the first third of the batches to 0.527 in the last. Over the same thirds, its mean score on real images, D(x), rose from 0.760 to 0.833. Its mean score on generated images, D(G(z)), fell from 0.140 to 0.067. It never dominated: its lowest epoch mean was 0.437, above F1's line of 0.425. The generator loss rose from 3.443 to 4.018. Its highest single batch, above 20, came in the first epoch, and later single batches spike

up to about 15. GANs "have been known to be unstable to train" (Radford et al., 2016, p. 1), and "Stabilization of GAN learning remains an open problem." (Goodfellow et al., 2016, p. 698). The discriminator kept its lead, and neither network collapsed.

Figure 3 follows the same 16 noise vectors through every checkpoint. They give a grey before training, a fine noise texture at epoch 1, a faint outline at epoch 5 and blurred orange shapes at epoch 10. From epoch 20 each vector gives a recognisable pinned specimen, and from epoch 40 each keeps much the same one. F4's diversity ratio was 0.53 at epoch 10, and between 0.93 and 1.03 from epoch 20 on.

Figure 3

Run 1's EMA Generator on the Same 16 Noise Vectors at Every Checkpoint



Note. From notebook Figure 7. One row per checkpoint, from before training at the top to epoch 100 at the bottom, each labelled with its diversity ratio (1.0 is as varied as 64 real training images). Generated by a model, not specimens.

Runs 2 and 3 show how much the stabilisers changed. Run 2, with spectral normalisation, still let the discriminator lead, ending with a mean D(x) of 0.833 and D(G(z)) of 0.067, the same as run 1's last third, by coincidence. Run 3, with TTUR as well, was the most even. Its discriminator began above the constant-0.5 reference, at 2.436 in epoch 1, and eased to 0.812 by epoch 100. But it learnt slowly: its live generator's diversity ratio was 0.65 at epoch 5, against 1.01 and 0.99 in runs 1 and 2. All three runs took 21.2 minutes on the graphics processing unit (GPU). On the central processing unit (CPU), I estimate 4.7 hours, from 20 timed steps of 0.37 seconds on 16 threads.

In plain terms, the networks trained as a GAN should. The generator went from grey noise to butterflies within 20 epochs and then held a broadly stable appearance, while the discriminator stayed ahead of it.

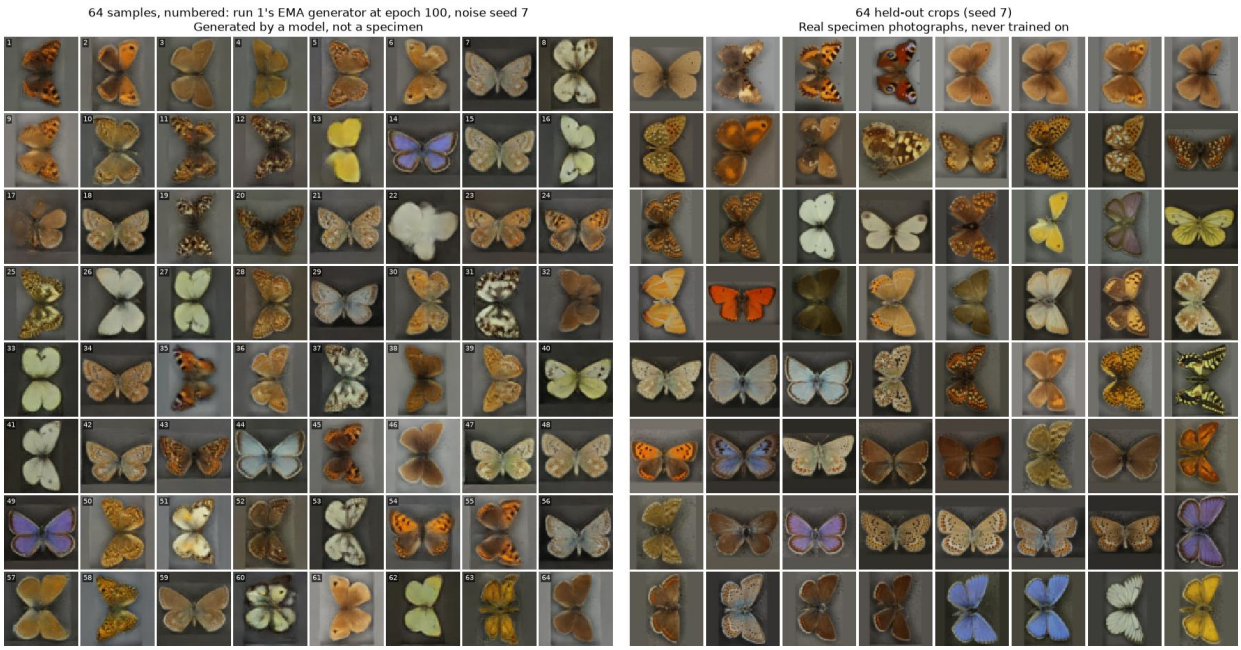
Output Evaluation and Interpretation

How I Evaluated. The core is a qualitative judgement against written criteria. "A subjective evaluation based on visual fidelity of samples is still clearly appropriate when the goal is image synthesis." (Theis et al., 2016, p. 8). I drew 64 samples from 64 noise vectors with seed 7 and numbered them 1 to 64. Figure 4 sets them beside 64 held-out crops, real photographs the model never saw. I cite each sample by its number, as #22. Every sample was scored from 0 to 1, in steps of 0.5, on five criteria fixed before training: wing symmetry, pattern coherence, body and antennae, background or label artefacts, and species plausibility. An AI assistant drafted a score and a reason for each of the 64 samples against these criteria. I checked and confirmed them, so the judgement is mine, as one lay rater, me, who knew the samples were generated. Ideally raters "who do not know the source of the samples" do this (Goodfellow et al., 2016, p.

714). The scores are tied to these samples' pixels: on a rerun that draws other samples, the notebook warns and hides them.

Figure 4

The 64 Numbered Samples Beside 64 Held-Out Images



Note. From notebook Figure 8. Left: samples from run 1's EMA generator at epoch 100, noise seed 7, generated by a model, not specimens. Right: real held-out crops adapted from the Natural History Museum (n.d.-b), CC BY 4.0 (https://creativecommons.org/licenses/by/4.0).

The criterion totals, each out of 64, were 63.5 for background or label artefacts and 57.5 for wing symmetry. Species plausibility scored 57.0, pattern coherence 51.5, and body and antennae 44.0. The mean is 4.27 of 5. Of the 64, 24 score 5 of 5, and three score under 3: #17, #22 and #60.

Strengths.

- **The blues and browns are the most convincing.** #7, #14, #29 and #49 among the blues, and the brown argus-like #18 and #42, are symmetrical with a clear body, and each scores 5 of 5.
- **The samples cover the kinds of butterfly in the real grid, not just one.** They include blues (#14, #49), whites (#26, #41), yellows (#13, #62), plain browns (#59), fritillary-like speckled undersides (#50, #58) and a tortoiseshell-like pattern (#35).
- **The model reproduces the photography as well as the butterfly.** The grey tray and the paler platform under the specimen appear throughout. Every sample but #60 scores 1 for background or label artefacts, a clean tray as in the photographs.

Limitations.

- **Bodies and antennae are weak.** This is the lowest criterion, at 44.0 of 64. #13, #33 and #54 have no body between the wings, and antennae show on only a few samples, such as #28 and #52. At 64 by 64 pixels, a body is two or three pixels wide.
- **Fine patterns blur and do not mirror.** The marks of #1 blur, the mottling of #11 is noisy and the chequer of #31 is coarse. In #30 and #36 an eyespot shows on one half only, and in #16 the spots differ between the halves.
- **The samples are blends, not species.** The model has no species label and sometimes mixes traits, such as the cream wings with painted-on yellow smudges of #51. Species plausibility is my lay judgement.

Failure Cases.

- **#22 is a white blob** with no wings, no body and no pattern, and scores 0 on four criteria. Its three nearest training images in the memorisation check are all whites, so it reads as a failed white.
- **#60 blends pale wings into four lobes** under a brown diagonal, with a dark wing above them.
- **#17 is lopsided.** Its left wings fade into a dark ghost while its right wings are clear and brown.

Variability. Mode collapse is when "only few prototype samples are generated" (Heusel et al., 2017, p. 4). I see no obvious gross collapse in the 64 samples I inspected. Supplementary measure S1 compares colour histograms with all held-out images. The samples score 0.062, closer than 64 real training images, at 0.075. Supplementary measure S2, the mean pairwise distance among the 64 samples on the 0 to 255 pixel scale, is 4,956.8, against 4,968.2 among the 64 held-out images, a ratio of 1.00. So this draw has a similar mean pixel-space spread to real images. Neither measure shows how well the samples cover the species or families, which I did not measure. Both measures are labelled supplementary, because no source I hold proposes them, and "the choice of metric must match the intended use of the model" (Goodfellow et al., 2016, p. 715). Both read colour and spread only, not wing shape. I did not compute the Fréchet Inception Distance (FID). Heusel et al. (2017, p. 6) computed it by passing images "through the pretrained Inception-v3 model", and this project uses no pre-trained network.

Label and Edge Artefacts. About 6 per cent of the 600 crops scored by eye had failed, and every crop was used. So I looked for label traces in the samples. A border measure found no near-white strip on any sample, as on the held-out images. But it also found none on the 21

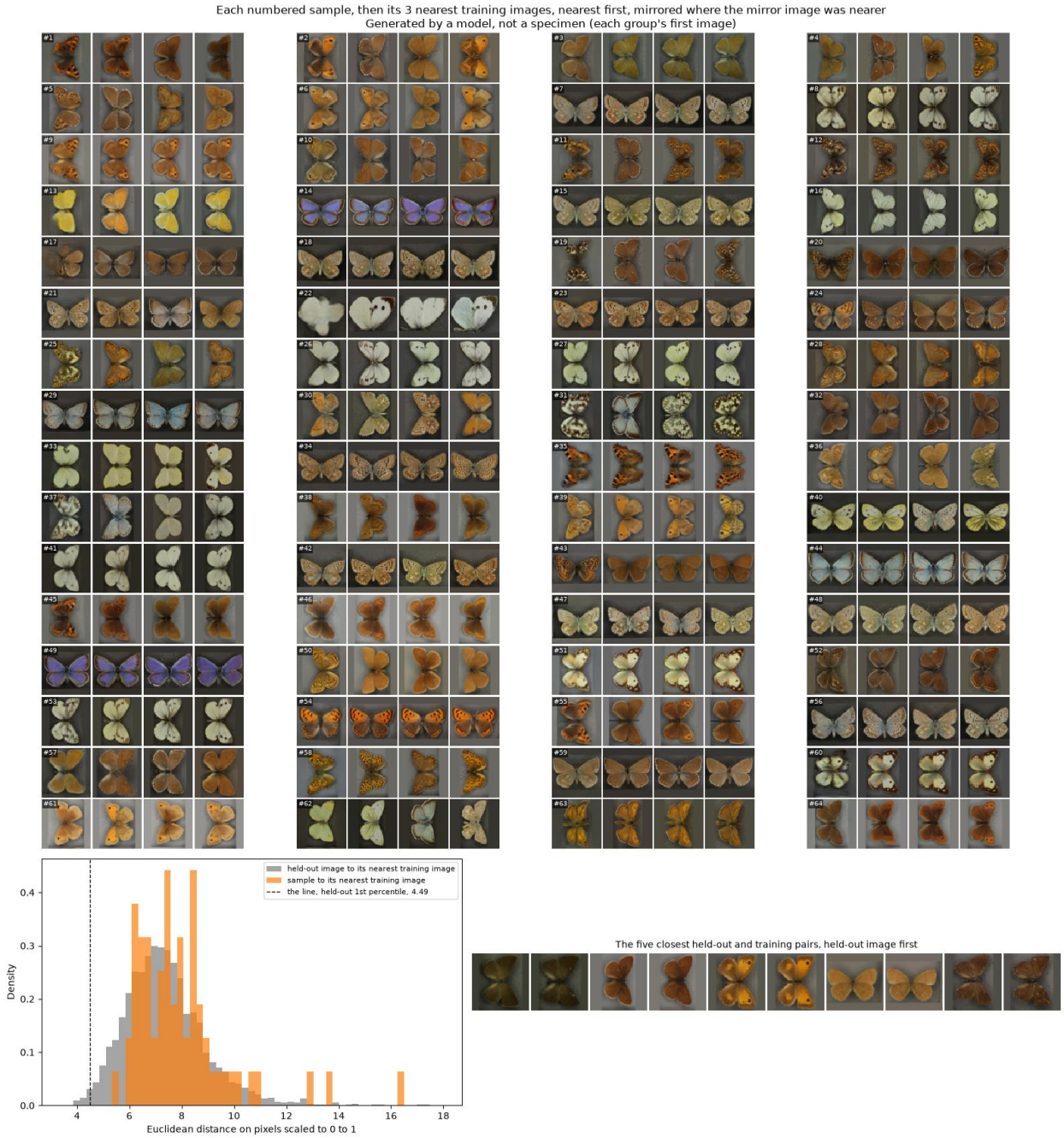
audited crops known to keep part of a label or card, so it cannot detect the traces it looks for. The evidence that stands is by eye: no sample shows a label block. The clearest possible trace is a faint dark mark near #60's bottom right corner. At this size I cannot tell it from a mark on the tray.

The Memorisation Check. I compared each sample with its 3 nearest training images, by Euclidean distance on the pixels. This follows the nearest neighbour (NN) search of Feng et al. (2021, p. 6704), who "find its NN in the corresponding training set with Euclidean distance in the original pixel space". The search includes each training image's mirror image, because training flips images. The line comes from real data. Somepalli et al. (2023, p. 6052) "draw random training images and compute the similarity with their closest match from the remaining training images" as a baseline. In the same spirit, I measured each held-out image against its nearest training image, and took the 1st percentile of those distances, 4.495, as the line.

Figure 5 shows each sample beside those 3 images, and below them the distances and the five closest held-out and training pairs. None of the 64 samples crossed this heuristic threshold. The nearest is #38, at 5.53. The samples' median distance, 7.63, sits close to the held-out images' median of 7.22. So the check finds no sample that is a near pixel-for-pixel copy of a training image or of its mirror image. Each of the five closest held-out and training pairs is two specimens of one species. Four of the five have close catalogue numbers, such as BMNH(E)1290870 and BMNH(E)1290871. I take it, from the numbers alone, that those were photographed in one session.

Figure 5

The Memorisation Check: Each Sample With Its 3 Nearest Training Images



Note. From notebook Figure 9. Crops adapted from the Natural History Museum (n.d.-b), CC BY 4.0 (https://creativecommons.org/licenses/by/4.0).

A positive control shows what the line catches. I made near-copies of 16 training images, each changed in ten ways: noise at three strengths, a blur at two radii, JPEG recompression at three qualities and a shift of one or two pixels. That gave 160 copies, and I searched for each as for a sample. The line caught every copy with Gaussian noise up to a standard deviation of 10 on the 0 to 255 scale, every blurred copy, and every JPEG copy at quality 90 and 75. At quality 50 it caught 15 of 16. Noise of 10 already sat close to the line, at a median of 4.36. A one-pixel shift was caught in 6 of 16 copies and a two-pixel shift in none. In all, it caught 133 of the 160 copies. So the line catches light pixel-level changes to a whole image and misses small shifts.

The check has limits. Theis et al. (2016, p. 6) "argue that it is unfit to detect any but the starkest forms of overfitting", and suggest "showing more than one nearest neighbor", which is why I show three. Distances on raw pixels are fragile: "a shift by only two pixels already caused a different nearest neighbor" (Theis et al., 2016, p. 6). They also warn that "A model which stores transformed training images can trivially pass the nearest-neighbor overfitting test" (Theis et al., 2016, p. 6), and the mirror search covers only one such transform. So a shifted copy, or a model that copied parts of images, could still pass, and the check cannot rule out memorisation in other forms.

In plain terms, the model learned what a British butterfly specimen on a museum tray looks like. It is weakest at fine detail: bodies, antennae and patterns that should match on both wings. None of the 64 samples crossed my copy check's threshold, but the check cannot rule out memorisation in other forms.

Ethical Considerations and Responsible Use

Collecting Bias. The data hold what collectors chose to collect and the Museum chose to keep. Paterson et al. (2016, p. 11) report "distinct 'hotspots' particularly along the south coast of

England". They also note that "British lepidopterists became interested, perhaps obsessed with variants and aberrations of species" (Paterson et al., 2016, p. 12). More widely, "societal preferences, rather than research activity, strongly correlate with taxonomic bias" in biodiversity data (Troudet et al., 2017, p. 1). My counts show the skew. Two families make up 81.4 per cent of the set, and the swallowtails 0.5 per cent. The model is unconditional, so it learns from the mix it was given, and I cannot count how faithfully its samples follow it. None of my 64 written reasons names a swallowtail. A generator like this would show users the common families of collections, not the butterflies of Britain as they live. That is also why I did not filter by the crop flag, which marks swallowtails more than four times as often as skippers.

Generated Specimens Passed Off as Evidence. A sample has no specimen, no locality and no date behind it. Yet 24 of the 64 samples score 5 of 5 on my rubric, so some could pass for specimens at this size. In UK rules, a specimen is "a living or dead animal or plant (or any of its parts or derivatives)" (Animal and Plant Health Agency & Department for Environment, Food & Rural Affairs, 2026). Under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), trade in some butterflies is regulated (Environment Canada, 2000). A guide for border officers notes that "Commercial shipments routinely use false declarations on import." (Environment Canada, 2000, "Why Are Butterflies Imported?" section). Where a trade already relies on false paperwork, a convincing photograph of a specimen that never existed could be used to back it. Every sample figure in my notebook carries the title "Generated by a model, not a specimen". The UK government's AI Playbook asks officials to "Clearly signpost when AI has been used to create content" (Government Digital Service, 2025, p. 51). No UK law requires such a label (Department for Science, Innovation and Technology & Department for Culture, Media and Sport, 2026, p. 12).

Memorisation and Creative Ownership. "Replicating training samples is also problematic for content creation due to potential copyright infringement" (Feng et al., 2021, p. 6701). For image generators, replication bears on "attributions to artists and photographers" (Somepalli et al., 2023, p. 6049). In the UK copyright report, memorisation risk "increases the smaller and more repetitive the dataset" (Department for Science, Innovation and Technology & Department for Culture, Media and Sport, 2026, p. 29). Here the photographers are the Museum's digitisers. None of the 64 samples crossed my memorisation check's heuristic threshold, the nearest at 5.53 against a line of 4.495. A positive control shows the line catches lightly changed copies of whole images. But it cannot rule out shifted copies (Theis et al., 2016, p. 6) or copied parts (Somepalli et al., 2023, p. 6049).

Attribution, Endorsement and Personal Data. The CC BY 4.0 licence lets me reuse the photographs if I credit the Museum. Its terms ask anyone who reuses the data to "explicitly state that the Museum cannot warrant the quality or accuracy of the data" (NHM, n.d.-c). They also forbid implying "the endorsement, affiliation, support or approval of the Museum" (NHM, n.d.-c). The UK copyright report records Wikimedia's view that attribution is "a core principle" of its open licences (Department for Science, Innovation and Technology & Department for Culture, Media and Sport, 2026, p. 27). A sample blends many photographs, so it cannot credit them one by one. I credit the Museum in the notebook's attribution lines, in this report's figure notes and in the Dataset section. The labels in each photograph carry collectors' handwriting and names, and "AI models themselves can contain personal data" (Information Commissioner's Office [ICO], 2026). The crop is set to leave the labels out, but 21 of the 600 crops scored by eye kept part of a label or card. The manifest keeps no field that names a person.

How This Could Be Used in the Real World. I propose three uses, ranked by the evidence. Each is my proposal. The first two are untested here, and the third is the idea I tested, which did not work.

1. **Images of rare species for identification tools.** Species photographs are uneven, as "the natural world is heavily imbalanced, as some species are more abundant and easier to photograph than others" (Van Horn et al., 2018, p. 8769). A GAN can be conditioned on a class label (Odena et al., 2017, p. 2642). A conditional version of my GAN could add images of rare species to a training set. Mine is unconditional and cannot choose a species.
2. **Material for spotting forged specimens.** Generated fakes could help train people or tools to tell a real specimen photograph from a made one. This sits close to the misuse above, and no source I hold shows it done.
3. **A quality flag for museum digitisation, tested here.** Digitisation projects run "quality control activities that are performed after digitisation" (Nieva de la Hidalga et al., 2019, p. 3). The NHM and the Royal Botanic Garden Edinburgh describe a tool for "detecting missing objects (such as colour reference cards or barcodes), identifying cropping errors" on herbarium sheets (Zhang et al., 2026). In factory inspection, models learn from defect-free images because "only a very limited amount of images with defects is available" (Bergmann et al., 2019, p. 9593). My idea was that the trained discriminator had learned what a normal specimen photograph looks like, so it could flag odd ones.

What the Test of the Quality Flag Showed. I tested the quality-flag idea once. Run 1's discriminator at epoch 100 scored 64 held-out images, 64 training images, the 64 samples and 34 known oddities. The oddities are the 34 crops that failed by eye in the crop audit before training:

21 kept part of a label or card, and 13 cut the specimen. If the idea worked, the oddities would score lower than the held-out images. They scored higher: a mean of 0.490 against 0.350. An oddity scored below a held-out image in only 30 per cent of pairs. The crops that cut the specimen scored highest, at 0.573. The samples scored 0.056 and the training images 0.457. So this discriminator tells generated images from real ones, and it tends to favour images it trained on. In this exploratory comparison it showed no useful separation of odd photographs from held-out ones. The comparison is weak on both sides. The 64 held-out images are a random draw, not checked as normal, and 30 of the 34 oddities were in training, with only 4 held out. A discriminator that favours its training images fits the "discriminator overfitting" that Karras et al. (2020, p. 12104) describe. The result is negative, and I keep it, because it shows the idea run and tested. With more or better data I would do four things differently. I would train a separate anomaly model only on crops checked as good. I would hold out labelled oddities for validation and never train on them. I would clean the crops first. I would add label-only photographs as a further oddity class, as none was on disk for this test.

Responsible Use. The model card idea asks a model's author to list "Out-of-scope uses", including "related contexts that users could try to apply the model to" (Mitchell et al., 2019, p. 223). This model's intended use is teaching and research: showing what a small GAN learns from a museum collection and where it fails. It is out of scope as a specimen record, as evidence in trade or research, and as training data for an identification tool without checks. My design choices follow from that. The crop is set to remove labels, and the manifest drops every field that can name a person. Each sample figure says it is generated, and this report names specimens only by catalogue number. The Museum's photographs are credited, with its no-warranty statement and no claim of its endorsement.

Limitations and Future Improvements

Limitations of the Model. Fine detail is the main weakness. Body and antennae scored 44.0 of 64 and pattern coherence 51.5. At 64 by 64 pixels, a body or an antenna is only a few pixels wide, so the resolution caps what the model can show. The model is unconditional, so it cannot be asked for a species, and some samples blend traits, as #51 does. The checkpoint rule's F3 line was set too tight for this generator, and F4 read the EMA before it had warmed up. So the model I evaluate was chosen after I had seen the outputs. Run 3 was the most even run but learnt slowly, and I did not test a longer schedule.

Limitations of the Data. The data carry the collectors' choices, with two families making up most of the set. About 6 per cent of the 600 crops scored by eye had failed, and all crops were used. The closest held-out and training pair is only 3.87 apart, two specimens of one species. Four of the five closest pairs have catalogue numbers within 121 of each other. So a split by specimen still leaves near twins on both sides, and the memorisation line is set by them. Every photograph follows one museum protocol, so the model knows nothing of living butterflies.

Limitations of the Evaluation. The scores were drafted with an AI assistant and confirmed by one lay rater, me, knowing they were generated. No expert has rated them, so I cannot say whether they would fool a lepidopterist. The memorisation check finds only near pixel-for-pixel copies of whole images, or of their mirror images, and cannot rule out memorisation in other forms. Its positive control shows it misses small shifts. S1 and S2 read colour and spread, not shape, and the 64 samples are a single draw.

Risks. Two risks come from using the outputs, apart from the ethical concerns above. If someone used the samples as training images for an identification tool, it would learn blends and

missing bodies, as in #13, #51 and #60. And a reader could take my memorisation result to mean the model copied nothing, which the check cannot show.

Questions the Data Cannot Answer. Would the samples fool an expert? I have no expert rating. Are rare species under-represented in the samples? The model does not know species, so I cannot count them, and the conditional GAN below is the route to that answer. Does the 6 per cent of failed crops show in the samples? The border measure cannot answer it, as it missed all 21 audited crops that kept part of a label, and #60's corner mark is the only possible trace I can see by eye. A larger draw could show more.

Next Steps. Each step answers a limitation above.

1. **Train a conditional GAN.** A GAN given the species or family as a label (Odena et al., 2017, p. 2642) would answer the species question. It would allow a count of samples per family, and it is the route to the rare-species use.
2. **Recalibrate the failure flags.** I would read F4 on the live generator, or warm the EMA up, and set F3's line on a short pilot run before the main run. That answers the misfire.
3. **Use blind raters, including an expert.** Mixed real and generated images, scored by people who do not know which is which (Goodfellow et al., 2016, p. 714), would answer the single-rater limit.
4. **Clean the crops and split by session.** Crops checked against the labels, and a held-out set split by runs of catalogue numbers, would answer the failed crops and the near twins.
5. **Build a dedicated anomaly model.** For the quality-flag use, I would train a model on good crops only and test it on held-out oddities, as the test of the quality flag showed.
6. **Train longer with run 3's settings.** Run 3's spectral normalisation and TTUR gave the most even training. A longer schedule would test whether its samples catch up.

References

Animal and Plant Health Agency & Department for Environment, Food & Rural Affairs. (2026, June 23). *Check if you need a CITES permit to import or export endangered species*. GOV.UK. https://www.gov.uk/guidance/cites-imports-and-exports

Bergmann, P., Fauser, M., Sattlegger, D., & Steger, C. (2019). MVTec AD: A comprehensive real-world dataset for unsupervised anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 9592-9600). Computer Vision Foundation. https://openaccess.thecvf.com/content_CVPR_2019/html/Bergmann_MVTec_AD_--_A_Comprehensive_Real-World_Dataset_for_Unsupervised_Anomaly_CVPR_2019_paper.html

Department for Science, Innovation and Technology & Department for Culture, Media and Sport. (2026). *Report on copyright and artificial intelligence*. GOV.UK. https://www.gov.uk/government/publications/report-and-impact-assessment-on-copyright-and-artificial-intelligence

Environment Canada. (2000). *CITES identification guide: Butterflies*. Minister of Supply and Services Canada. https://cites.unia.es/file.php/1/files/CAN-CITES_Butterfly_Guide.pdf

Feng, Q., Guo, C., Benitez-Quiroz, F., & Martinez, A. M. (2021). When do GANs replicate? On the choice of dataset size. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 6701-6710). Computer Vision Foundation. https://openaccess.thecvf.com/content/ICCV2021/html/Feng_When_Do_GANs_Replicate_On_the_Choice_of_Dataset_Size_ICCV_2021_paper.html

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press. https://www.deeplearningbook.org

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, & K. Weinberger (Eds.), *Advances in neural information processing systems* (Vol. 27). Curran Associates. https://papers.nips.cc/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html

Government Digital Service. (2025). *Artificial intelligence playbook for the UK government*. GOV.UK. https://www.gov.uk/government/publications/ai-playbook-for-the-uk-government

Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 30). Curran Associates. https://papers.nips.cc/paper/2017/hash/8a1d694707eb0fefe65871369074926d-Abstract.html

Information Commissioner's Office. (2026). *Data protection challenges and our expectations*. In *Building trust and transparency into generative AI development: Our work to create regulatory certainty*. https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/building-trust-and-transparency-into-generative-ai-development-our-work-to-create-regulatory-certainty/data-protection-challenges-and-our-expectations/

Karras, T., Aittala, M., Hellsten, J., Laine, S., Lehtinen, J., & Aila, T. (2020). Training generative adversarial networks with limited data. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems* (Vol. 33, pp. 12104-12114). Curran Associates. https://papers.nips.cc/paper/2020/hash/8d30aa96e72440759f74bd2306c1fa3d-Abstract.html

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 220-229). Association for Computing Machinery. https://doi.org/10.1145/3287560.3287596

Miyato, T., Kataoka, T., Koyama, M., & Yoshida, Y. (2018). *Spectral normalization for generative adversarial networks* [Conference paper]. International Conference on Learning Representations (ICLR 2018). https://openreview.net/forum?id=B1QRgziT-

Natural History Museum. (n.d.-a). *Citation and attribution*. Natural History Museum Data Portal. Retrieved October 13, 2026, from https://data.nhm.ac.uk/about/citation

Natural History Museum. (n.d.-b). *Collection specimens* [Data set]. Natural History Museum Data Portal. Retrieved October 13, 2026, from https://doi.org/10.5519/0002965

Natural History Museum. (n.d.-c). *Terms and conditions*. Natural History Museum Data Portal. Retrieved October 13, 2026, from https://data.nhm.ac.uk/terms-conditions

Nieva de la Hidalga, A., van Walsun, M., Rosin, P., Sun, X., & Wijers, A. (2019). *Quality management methodologies for digitisation operations* (ICEDIG Deliverable D3.1) [Report]. Zenodo. https://doi.org/10.5281/zenodo.3469521

Odena, A., Olah, C., & Shlens, J. (2017). Conditional image synthesis with auxiliary classifier GANs. In *Proceedings of the 34th International Conference on Machine Learning* (PMLR 70, pp. 2642-2651). https://proceedings.mlr.press/v70/odena17a.html

Paterson, G., Albuquerque, S., Blagoderov, V., Brooks, S., Cafferty, S., Cane, E., Carter, V., Chainey, J., Crowther, R., Douglas, L., Durant, J., Duffell, L., Hine, A., Honey, M., Huertas, B., Howard, T., Huxley, R., Kitching, I., Ledger, S., . . . Wing, P. (2016). iCollections: Digitising the British and Irish butterflies in the Natural History Museum, London. *Biodiversity Data Journal, 4*, Article e9559. https://doi.org/10.3897/BDJ.4.e9559

Radford, A., Metz, L., & Chintala, S. (2016). *Unsupervised representation learning with deep convolutional generative adversarial networks* [Conference paper]. International Conference on Learning Representations (ICLR 2016), San Juan, Puerto Rico. https://doi.org/10.48550/arXiv.1511.06434

Salimans, T., Goodfellow, I., Zaremba, W., Cheung, V., Radford, A., & Chen, X. (2016). Improved techniques for training GANs. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 29). Curran Associates. https://papers.nips.cc/paper/2016/hash/8a3363abe792db2d8761d6403605aeb7-Abstract.html

Somepalli, G., Singla, V., Goldblum, M., Geiping, J., & Goldstein, T. (2023). Diffusion art or digital forgery? Investigating data replication in diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 6048-6058). Computer Vision Foundation.

https://openaccess.thecvf.com/content/CVPR2023/html/Somepalli_Diffusion_Art_or_Digital_Forgery_Investigating_Data_Replication_in_Diffusion_CVPR_2023_paper.html

Theis, L., van den Oord, A., & Bethge, M. (2016). *A note on the evaluation of generative models* [Conference paper]. International Conference on Learning Representations (ICLR 2016), San Juan, Puerto Rico. https://doi.org/10.48550/arXiv.1511.01844

Troudet, J., Grandcolas, P., Blin, A., Vignes-Lebbe, R., & Legendre, F. (2017). Taxonomic bias in biodiversity data and societal preferences. *Scientific Reports, 7*, Article 9132. https://doi.org/10.1038/s41598-017-09084-6

Van den Broeck, J., Argeseanu Cunningham, S., Eeckels, R., & Herbst, K. (2005). Data cleaning: Detecting, diagnosing, and editing data abnormalities. *PLoS Medicine, 2*(10), Article e267. https://doi.org/10.1371/journal.pmed.0020267

Van Horn, G., Mac Aodha, O., Song, Y., Cui, Y., Sun, C., Shepard, A., Adam, H., Perona, P., & Belongie, S. (2018). The iNaturalist species classification and detection dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 8769-8778). Computer Vision Foundation. https://openaccess.thecvf.com/content_cvpr_2018/html/Van_Horn_The_INaturalist_Species_CVPR_2018_paper.html

Zhang, N., Salili-James, A., Poon, S. T. S., Hartley, B., Drinkwater, R., Phillips, M., Cubey, R., & Haston, E. M. (2026). AI-assisted quality checks for herbarium sheet digitisation. *Biodiversity Information Science and Standards, 10*, Article e184058. https://doi.org/10.3897/biss.10.184058